

# Simultaneous equations and inequalities - Set 2 - Solutions

Specification MM1.4, MM1.5, M4.16, M4.17.

## Answer key

| Question | Answer | Topic                                   |
|----------|--------|-----------------------------------------|
| 1        | B      | Simultaneous equations and inequalities |
| 2        | A      | Simultaneous equations and inequalities |
| 3        | H      | Simultaneous equations and inequalities |
| 4        | E      | Simultaneous equations and inequalities |
| 5        | F      | Simultaneous equations and inequalities |
| 6        | C      | Simultaneous equations and inequalities |
| 7        | B      | Simultaneous equations and inequalities |

## Worked solutions

### Question 1 SIM-08 · Simultaneous equations and inequalities · spec MM1.5

Which of the following inequalities has exactly the same solution set as  $\frac{1}{x} > 2$ ?

**Answer: B**

If  $x > 0$ :  $1 > 2x$ , so  $0 < x < \frac{1}{2}$ . If  $x < 0$ :  $\frac{1}{x}$  is negative and cannot exceed 2. So  $0 < x < \frac{1}{2}$ . (Multiplying by  $x$  without considering its sign is the classic error.)

#### Full method - how and why

An inequality with the variable in the denominator. You must not multiply by  $x$  without knowing its sign - split into cases instead.

- 1 Case  $x > 0$ : multiplying by  $x$  keeps the direction, giving  $1 > 2x$ , i.e.  $x < \frac{1}{2}$ . Combined with  $x > 0$ :  $0 < x < \frac{1}{2}$ .  
why: Multiplying an inequality by a positive quantity is safe; the case assumption must be carried into the answer.
- 2 Case  $x < 0$ : then  $\frac{1}{x} < 0 < 2$ , so the inequality can never hold.  
why: Rather than multiplying by a negative (and flipping), just observe the sign - a negative number is never greater than 2.
- 3 Case  $x = 0$ :  $\frac{1}{x}$  is undefined, so  $x = 0$  is excluded.  
why: Always check the value that breaks the expression.
- 4 Combine: the solution set is exactly  $0 < x < \frac{1}{2}$ .  
why: Only the first case contributes.

#### Where it goes wrong

- Multiplying through by  $x$  blindly to get  $x < \frac{1}{2}$ , which wrongly includes every negative number.
- Answering  $x > 2$  by 'inverting' both sides without justification.

**Takeaway** Never multiply an inequality by an expression of unknown sign: split into cases, or bring everything to one side and analyse the sign of the fraction.

### Question 2 SIM-10 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of  $x$  for which  $(x - 1)(x - 3)(x - 5) > 0$ .

**Answer: A**

Sign chart: the product is positive for  $x > 5$ , negative for  $3 < x < 5$ , positive for  $1 < x < 3$ , negative for  $x < 1$ .

#### Full method - how and why

A cubic inequality. With three distinct linear factors, use a sign chart - the sign alternates as you cross each root.

- 1 The roots are  $x = 1, 3, 5$ ; mark them on a number line, splitting it into four regions.  
why: A product of linear factors can only change sign at a root.
- 2 Test the far-right region, e.g.  $x = 6$ :  $(5)(3)(1) > 0$ .  
why: One test value fixes the whole chart; the far right is easiest because every factor is positive.
- 3 Alternate leftwards: positive for  $x > 5$ , negative on  $(3, 5)$ , positive on  $(1, 3)$ , negative for  $x < 1$ .  
why: Each simple (unrepeated) root flips the sign exactly once.
- 4 Collect the positive regions:  $1 < x < 3$  or  $x > 5$ .  
why: The answer is a union of two intervals.

#### Where it goes wrong

- Answering  $x > 5$  only, forgetting the middle positive region.
- Assuming the expression is positive whenever  $x$  is bigger than all the roots - true here, but the alternation is what you must use.

**Takeaway** Sign chart: find the roots, test one region, then alternate; repeated factors are the only ones that do not flip the sign.

**Question 3** SIM-11 · Simultaneous equations and inequalities · spec MM1.5

Given that  $|2x - 3| < 5$ , which of the following statements is/are true?

**I**  $-1 < x < 4$

**II**  $x^2 < 16$

**III**  $x > -1$

**Answer: H**

$-5 < 2x - 3 < 5 \Rightarrow -1 < x < 4$ , so I is true and III follows from I. For II: every  $x$  in  $(-1, 4)$  has  $x^2 < 16$  (the largest possible  $|x|$  is just under 4), so II is also true.

**Full method - how and why**

A modulus inequality plus three consequences. Solve the modulus first, then test each statement against that interval.

**1** Unpack the modulus:  $|2x - 3| < 5$  means  $-5 < 2x - 3 < 5$ .

*why:*  $|A| < k$  means  $A$  lies within  $k$  of zero - a double inequality, not two separate cases.

**2** Solve the double inequality: add 3 throughout to get  $-2 < 2x < 8$ , then halve:  $-1 < x < 4$ . So **I is true**.

*why:* Whatever you do to one part of a double inequality you do to all three parts.

**3** III says  $x > -1$ , which is half of I. **True**.

*why:* A weaker statement implied by a true one is automatically true.

**4** II: for  $-1 < x < 4$ , the largest  $|x|$  can get is just under 4, so  $x^2 < 16$  always. **True**.

*why:* For squares, what matters is the largest **absolute value** in the interval, not the largest value.

**5** All three hold.

*why:* Match to the option list.

**Where it goes wrong**

- Treating  $|2x - 3| < 5$  as  $2x - 3 < 5$  only, losing the lower bound.
- Rejecting II by testing a value like  $x = -3$  that is not in the interval at all.

**Takeaway**  $|A| < k \Leftrightarrow -k < A < k$ ; when squaring an interval, check the end with the larger absolute value.

**Question 4** SIM-13 · Simultaneous equations and inequalities · spec MM1.5

How many integers  $x$  satisfy  $x^2 < 5x + 24$  and  $x^2 > 3x$ ?

**Answer: E**

$x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$ .  $x(x - 3) > 0 \Rightarrow x < 0$  or  $x > 3$ . Integers:  $-2, -1$  and  $4, 5, 6, 7$ : six.

**Full method - how and why**

Two quadratic inequalities, then an integer count. The second one factorises without a constant term, which changes its shape.

- 1 First:  $x^2 - 5x - 24 < 0 \Rightarrow (x - 8)(x + 3) < 0 \Rightarrow -3 < x < 8$ .  
why: Between the roots for  $< 0$ .
- 2 Second:  $x^2 - 3x > 0 \Rightarrow x(x - 3) > 0 \Rightarrow x < 0$  or  $x > 3$ .  
why: Roots 0 and 3; outside the roots for  $> 0$ . Do not divide by  $x$  - that would lose the  $x < 0$  branch.
- 3 Intersect: from  $(-3, 8)$  keep  $(-3, 0)$  and  $(3, 8)$ .  
why: Overlap both solution sets on one number line.
- 4 Count integers:  $-2, -1$  from the first piece and  $4, 5, 6, 7$  from the second: six.  
why: Endpoints  $-3, 0, 3, 8$  are all excluded by strict inequalities.

**Where it goes wrong**

- Dividing  $x^2 > 3x$  by  $x$  to get  $x > 3$ , losing all the negative solutions.
- Including 0 or 3 in the count.

**Takeaway** Never divide an inequality by the variable; factorise instead, then count integers explicitly.

**Question 5** SIM-15 · Simultaneous equations and inequalities · spec MM1.5

The solution set of  $ax^2 + bx + c < 0$  (with  $a > 0$ ) is  $-2 < x < 5$ . What is the solution set of  $ax^2 - bx + c < 0$ ?

**Answer: F**

Replacing  $x$  by  $-x$  reflects the graph in the  $y$ -axis, so the roots  $-2$  and  $5$  become  $2$  and  $-5$ :  $-5 < x < 2$ .

**Full method - how and why**

A transformation question hidden in an inequality. Replacing  $x$  by  $-x$  reflects the graph in the  $y$ -axis, and the solution set reflects with it.

- 1 Note that  $a(-x)^2 + b(-x) + c = ax^2 - bx + c$ ; the second expression is the first with  $x$  replaced by  $-x$ .  
why: Spotting this makes the question a one-liner; the alternative is to reconstruct  $a, b, c$  explicitly.
- 2 So the graph of the second is the reflection of the first in the  $y$ -axis.  
why:  $f(-x)$  is always the  $y$ -axis reflection of  $f(x)$ .
- 3 The original is negative on  $-2 < x < 5$ ; reflecting sends  $x$  to  $-x$ , so the roots  $-2$  and  $5$  move to  $2$  and  $-5$ .  
why: Every point of the solution set flips sign, and the interval flips with it.
- 4 The new solution set is  $-5 < x < 2$ .  
why: Write the interval with the smaller number first.

**Where it goes wrong**

- Leaving the interval as  $2 < x < -5$ , which is empty as written.
- Assuming the solution set is unchanged because 'only the sign of  $b$  changed'.

**Takeaway** Changing  $b$  to  $-b$  reflects a quadratic in the  $y$ -axis; everything about the graph, including solution sets, reflects too.

**Question 6** SIM-16 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of  $x$  for which  $\frac{x-2}{x+1} \leq 0$ .

**Answer: C**

A fraction is  $\leq 0$  when numerator and denominator have opposite signs (or the numerator is 0):  $x - 2 \leq 0$  and  $x + 1 > 0$  gives  $-1 < x \leq 2$ ;  $x - 2 \geq 0$  and  $x + 1 < 0$  is impossible.  $x = -1$  is excluded as the fraction is undefined.

**Full method - how and why**

A rational inequality. Never multiply by  $x + 1$  (its sign is unknown) - analyse the sign of the fraction instead.

- 1 Find the critical values: numerator zero at  $x = 2$ , denominator zero at  $x = -1$ .

why: The fraction can only change sign where the numerator or denominator vanishes.

- 2 Test the three regions. For  $x < -1$  (say  $x = -2$ ):  $\frac{-4}{-1} = 4 > 0$ . For  $-1 < x < 2$  (say  $x = 0$ ):  $\frac{-2}{1} = -2 < 0$ . For  $x > 2$  (say  $x = 3$ ):  $\frac{1}{4} > 0$ .

why: One test value per region settles the sign throughout that region.

- 3 Include  $x = 2$  (the fraction is 0, and  $\leq 0$  allows it) but exclude  $x = -1$  (undefined).

why: The two endpoints behave differently: a zero numerator is allowed by  $\leq$ ; a zero denominator never is.

- 4 Answer:  $-1 < x \leq 2$ .

why: Note the mixed brackets - strict at  $-1$ , inclusive at 2.

**Where it goes wrong**

- Multiplying by  $x + 1$  to get  $x - 2 \leq 0$ , i.e.  $x \leq 2$ , which wrongly includes everything below  $-1$ .
- Writing  $-1 \leq x \leq 2$  and including a value where the expression is undefined.

**Takeaway** For  $\frac{A}{B} \leq 0$ , use a sign chart on  $A$  and  $B$ ; a zero denominator is always excluded, a zero numerator is included when the inequality is not strict.

**Question 7** SIM-18 · Simultaneous equations and inequalities · spec MM1.5

Consider the statement: for all real  $x$ , if  $x > 2$  then  $x^2 - 3x + 2 > 0$ . Which of the following is true?

**Answer: B**

$x^2 - 3x + 2 = (x - 1)(x - 2) > 0$  exactly when  $x < 1$  or  $x > 2$ . So  $x > 2$  does imply it (true), but the converse fails, e.g.  $x = 0$  gives  $2 > 0$  without  $x > 2$ .

**Full method - how and why**

A Paper 2-style wrapper on a quadratic inequality: decide the statement and its converse separately.

- 1 Factorise the quadratic:  $x^2 - 3x + 2 = (x - 1)(x - 2)$ , which is positive exactly when  $x < 1$  or  $x > 2$ .

why: Knowing precisely where the expression is positive lets you test both directions quickly.

- 2 The statement: does  $x > 2$  guarantee positivity? Yes -  $x > 2$  lies inside the region  $x > 2$ . **True.**

why: An implication is true when the hypothesis region sits inside the conclusion region.

- 3 The converse: does positivity guarantee  $x > 2$ ? No -  $x = 0$  gives  $(0 - 1)(0 - 2) = 2 > 0$  but  $0 \not> 2$ . **False.**

why: The positivity region is strictly larger than  $x > 2$ , so a counterexample must come from the extra part  $x < 1$ .

- 4 So the statement is true but its converse is false.

why: Report both verdicts in the order the options use them.

**Where it goes wrong**

- Assuming the converse holds because the implication does - converses are logically independent.
- Only testing values greater than 2, which can never refute the converse.

**Takeaway** An implication is true when one region is contained in the other; its converse needs the reverse containment, so look for a counterexample in the leftover region.